1 **(a)** 33 **[B1]**

(b) $7n + k$ for any integer k **[M1]**

$7n - 2$ **[A1]**

(c) $7n - 2 = 150$

$7\text{n} = 152 whichdoesnotgiveanintegervaluefor\text{n}.[B1]$

[Total Marks: 4]

2 **(a)** 29 **[B1]**

(b) $6n + c$ or finding common difference of 6 **[M1]**

$6n - 1$ **[A1]**

(c) Sets $6n - 1 = 150 \implies 6n = 151$ and states that 151 is not divisible by 6 (or n is not an integer) **[B1]**

[Total Marks: 4]

3 **(a)** Common difference is 4 **[M1]**

$4n + 1$ **[A1]**

(b) Identifies common ratio is 3 **[M1]**

$2 \times 3^{n-1}$ **[A1]**

[Total Marks: 4]

4 **(a)** $23, 27$ **[B1]**

(b) $4n + k$ (for any k) **[M1]**

$4n + 3$ **[A1]**

(c) Equates $4n + 3 = 102$ and attempts to solve for n **[M1]**

$n = 24.75$, which is not a whole number/integer **[A1]**

[Total Marks: 5]

5 **(a)** **(i)** 21 and 25 **[B1]**

(ii) Identifies common difference of 4 leading to $4n$ **[M1]**

$4n + 1$ **[A1]**

(b) Identifies common ratio of 3 **[M1]**

$2 \times 3^{n-1}$ **[A1]**

[Total Marks: 5]

6 **(a)** 17 **[B1]**

(b) $4n + c$ where c is a constant **[M1]**

$4n + 1$ **[A1]**

(c) $4n + 1 = 141$ or $141 - 1 = 140$ **[M1]**

35 **[A1]**

[Total Marks: 5]

7 **(a)** Identifying the second differences are constant 4 **[M1]**

Deducing the n^2 coefficient is 2, or setting up $2n^2 + bn + c$ **[M1]**

$2n^2 - n - 2$ **[A1]**

(b) $2n^2 - n - 2 = 433$ **[M1]**

$2n^2 - n - 435 = 0$ leading to correct factorisation $(2n + 29)(n - 15) = 0$ or correct substitution into the quadratic formula **[M1]**

15 (must reject negative solution if given) **[A1]**

[Total Marks: 6]

8 **(a)** First differences: $10, 20, 34, 52$ **[M1]**

Second differences: $10, 14, 18$
Third differences: $4, 4$
Since the third difference is a non-zero constant, the sequence is cubic. **[A1]**

(b) Uses $6a = 4 \implies a = \dfrac{2}{3}$ **[M1]**

Uses $12a + 2b = 10 \implies 12(\dfrac{2}{3}) + 2b = 10 \implies 8 + 2b = 10 \implies b = 1$ **[M1]**

Uses $7a + 3b + c = 10 \implies 7(\dfrac{2}{3}) + 3(1) + c = 10 \implies c = 10 - 3 - \dfrac{14}{3} = \dfrac{7}{3}$ **[M1]**

Uses $a + b + c + d = 4 \implies \dfrac{2}{3} + 1 + \dfrac{7}{3} + d = 4 \implies 4 + d = 4 \implies d = 0$ **[M1]**

$\dfrac{2}{3}n^3 + n^2 + \dfrac{7}{3}n$ **[A1]**

[Total Marks: 7]

9 **(a)** 79 **[B1]**

(b) Calculates second differences to be 4, or equates $2a = 4$ **[M1]**

Deduces that the sequence contains $2n^2$ **[M1]**

$2n^2 + n + 1$ **[A1]**

[Total Marks: 4]

10 **(a)** One correct term evaluated **[M1]**

$3, 3, 5$ **[A1]**

(b) Second differences are 4 **[M1]**

Identifies $2n^2$ as part of the expression **[M1]**

$2n^2 + n - 1$ **[A1]**

[Total Marks: 5]

11 (a) Finds first differences of $7, 13, 19$ and constant second differences of 6 **[M1]**

Deduces the quadratic term is $3n^2$ **[M1]**

$3n^2 - 2n + 1$ **[A1]**

(b) 35 **[B1]**

(c) $2^n + 3 = 131$ leading to $2^n = 128$ **[M1]**

7 **[A1]**

[Total Marks: 6]

12 (a) Finds first differences: $3, 7, 11$ and second differences: $4, 4$ **[M1]**

Deduces the term involves $2n^2$ **[M1]**

$2n^2 - 3$ **[A1]**

(b) $3^n - 2 \approx 1000 \implies 3^n \approx 1002$ **[M1]**

Uses logarithms or trial and improvement: $n = 6 \implies 3^6 - 2 = 729 - 2 = 727$;
$n = 7 \implies 3^7 - 2 = 2187 - 2 = 2185$ **[M1]**

727 **[A1]**

[Total Marks: 6]

13 (a) 77 **[B1]**

(b) Finds first differences $(7, 11, 15, 19)$ and constant second difference of 4 **[M1]**

Deduces the coefficient of n^2 is 2 **[A1]**

Subtracts $2n^2$ from the sequence terms to identify the linear sequence $n - 1$ **[M1]**

$2n^2 + n - 1$ **[A1]**

(c) Sets up the equation $2k^2 + k - 1 = 464$ **[M1]**

Rearranges to form a quadratic equal to zero: $2k^2 + k - 465 = 0$ **[M1]**

15 **[A1]**

[Total Marks: 7]

14 (a) Recognises the relationship to cube numbers $(1, 8, 27, 64, 125)$ **[M1]**

$n^3 + 1$ **[A1]**

(b) Identifies second difference is 2, or relates to square numbers **[M1]**

$n^2 + 2n$ or $(n+1)^2 - 1$ **[A1]**

(c) $\dfrac{n^3+1}{n^2+2n}$ **[B1]**

(d) Substitutes $n = 12$ into their expression for sequence R, yielding $\dfrac{1728+1}{144+24}$ or $\dfrac{1729}{168}$ **[M1]**

Simplifies the fraction correctly to $\dfrac{247}{24}$ **[A1]**

[Total Marks: 7]

15 (a) Finds third differences are 6 (first diffs: $9, 25, 47, 75$; second diffs: $16, 22, 28$; third diffs: 6) **[M1]**

Identifies n^3 as the cubic term **[B1]**

Subtracts n^3 from the sequence to get a quadratic sequence $(-2, 0, 6, 16, 30)$ and finds its second differences are 4 to give $2n^2$ **[M1]**

$n^3 + 2n^2 - 4n$ **[A1]**

(b) $5 \times 3^{k-1} = 3645$

$3^{k-1} = 729$ [M1]

7 **[A1]**

(c) $\dfrac{2}{3}$ or 0.667 **[B1]**

[Total Marks: 7]

16 (a) Finds first differences $14, 38, 74$ and second differences $24, 36$ **[M1]**

Finds third difference of 12 leading to a cubic term of $2n^3$ **[M1]**

Evaluates the sequence of residuals $T_n - 2n^3$ to find a constant difference of -1 **[M1]**

$2n^3 - 1$ **[A1]**

(b) (i) $V_1 = 1$, $V_2 = \dfrac{5}{4}$, $V_3 = \dfrac{5}{4}$ **[B2]**

Award B1 if any two are correct **[B1]**

(ii) Substitutes $n = 5$ to find $V_5 = \dfrac{25+1}{32} = \dfrac{13}{16}$ **[B1]**

Substitutes $n = 4$ to find $V_4 = \dfrac{16+1}{16} = \dfrac{17}{16}$ **[B1]**

Evaluates $\dfrac{13}{16} - \dfrac{17}{16} = -\dfrac{1}{4}$ (accept -0.25) **[A1]**

[Total Marks: 9]

17 (a) Correct first differences $1, 9, 23, 43$ and second differences $8, 14, 20$ and third differences 6 **[M1]**

Deducing the cubic term coefficient is $a = 6 \div 6 = 1$ **[B1]**

Setting up equations to solve for other coefficients, e.g. $12a + 2b = 8$ **[M1]**

$n^3 - 2n^2 + 5$ **[A1]**

(b) $2a + b = 10$ **[B1]**

$8a + b = 34$ **[B1]**

Correct method to solve simultaneous equations, e.g. subtracting to get $6a = 24$ **[M1]**

$a = 4, b = 2$ **[A1]**

(c) Substituting $n = 12$ into their $4 \times 2^n + 2$ **[M1]**

16386 **[A1]**

[Total Marks: 10]

18 (a) $4n + 1$ **[B2]**

If $4n + k$ seen, award **[M1]**

(b) (i) 182 **[B1]**

(ii) Identifies third differences are constant (6), so term contains n^3 **[M1]**

Subtracts n^3 terms $(1, 8, 27, 64, 125)$ from the sequence **[M1]**

Remainder sequence is $1, -2, -7, -14, -23$ **[M1]**

Deduces the remainder is $-n^2 + 2$ **[M1]**

$n^3 - n^2 + 2$ **[A1]**

(c) (i) $\dfrac{25}{182}$ **[B1]**

(ii) Recognises the pattern is Sequence A divided by Sequence B **[M1]**

$\dfrac{4n + 1}{n^3 - n^2 + 2}$ **[A1]**

[Total Marks: 10]